

Attention Sinks in Massively Multilingual Neural Machine Translation: Discovery, Analysis, and Mitigation

Hillary Mutisya
Thiomi NLP

John Mugane
Harvard University

Abstract

Cross-attention patterns in neural machine translation (NMT) are widely used as a window into how multilingual models process and align linguistic structure. In this work, we report a systematic artifact in cross-attention analysis of NLLB-200, Meta’s 600-million-parameter massively multilingual NMT model: non-content tokens—dominated by the end-of-sequence token $\langle /s \rangle$, with additional contributions from language identifier tags and punctuation—capture between 83 and 91 percent of total cross-attention mass. We term these concentrations *attention sinks*, extending prior findings on attention sink phenomena in large language models [Xiao et al., 2023] to the cross-attention mechanism of multilingual NMT and identifying a distinct causal mechanism rooted in tokenization and vocabulary design rather than position-based bias. We demonstrate that this artifact causes raw similarity metrics to underestimate content-level similarity by nearly a factor of two (36.7% raw vs. 70.7% filtered), rendering uncorrected cross-attention analyses unreliable. To address this, we develop and validate a *content-only filtering* methodology¹ that removes non-content tokens and renormalizes the remaining attention distribution. Applying this methodology to 1,000 parallel sentences across four African languages (Swahili, Kikuyu, Somali, Luo) and validating on four non-African languages (German, Turkish, Chinese, Hindi) spanning seven language families and three scripts, we confirm the artifact is universal and recover substantive linguistic signals previously masked: a 16.9 percentage-point gap between teacher-forcing and generation modes, clear language-family clustering in attention entropy and peak patterns, and a previously hidden Somali paradox linking SOV word order to monotonic alignment strategy. We release our filtering toolkit and corrected attention datasets to support reproducible interpretability research on multilingual NMT.

1. Introduction

The interpretability of multilingual neural machine translation models has become a pressing research priority. As models such as NLLB-200 [NLLB Team et al., 2022] scale to cover 200 languages simultaneously, understanding *how* they process and transfer linguistic knowledge across typologically diverse languages becomes essential for diagnosing failure modes, improving low-resource performance, and building theoretical accounts of cross-lingual generalization.

A dominant methodology in NMT interpretability is the analysis of *cross-attention patterns*: the weights with which decoder states attend to encoder representations at each decoding step. These weights have been treated as proxies for word alignment [Bahdanau et al., 2015, Raganato and Tiedemann, 2018], structural correspondences [Voita et al., 2019], and information routing across language pairs. Visualization tools such as BertViz [Fig, 2019] have made cross-attention analysis accessible, and numerous studies have used these patterns to draw conclusions about how multilingual models encode typological properties.

Our work began as a straightforward interpretability study of NLLB-200’s cross-attention patterns for four African languages—Swahili, Kikuyu, Somali, and Luo—representing the Bantu, Cushitic, and Nilotic language families respectively. During initial analysis of cross-attention heatmaps

for English-to-Swahili translation, we observed an anomaly: the vast majority of attention mass was concentrated not on content-bearing words but on a small set of non-content tokens. Language identifier tags such as `swh_Latn` and `eng_Latn`, punctuation marks, and structural special tokens ($\langle /s \rangle$, $\langle s \rangle$) collectively captured 83–91% of cross-attention weight across all four languages.

This concentration constitutes an *attention sink*: a token or small set of tokens that absorbs disproportionate attention not because of semantic relevance to the decoding target, but because of structural properties of the model or tokenization. Xiao et al. [2023] identified attention sinks in large autoregressive language models, where the initial token acts as a sink for attention that has no strong content-based destination. Our finding extends this phenomenon to the cross-attention mechanism of multilingual NMT, but with a distinct mechanism: rather than position-based bias, the sinks arise from special vocabulary items that are present in every sentence (language tags) or statistically ubiquitous (punctuation), giving them persistent, sentence-independent high attention.

The practical consequence is severe. Raw cross-attention similarity metrics computed without filtering are approximately *half* the content-only values: teacher-forcing similarity rises from 36.7% to 70.7% after filtering, a relative increase of $(70.7 - 36.7)/36.7 = 93\%$. Equivalently, the raw metric

underestimates the true content-level similarity by nearly a factor of two, because sink tokens concentrate attention mass away from content tokens, compressing all content-based differences into a narrow residual band. We make the following contributions:

1. **Discovery and characterization** of attention sinks in NLLB-200 cross-attention, with a breakdown by token type and language.
2. **Content-only filtering methodology**: a principled pipeline for removing non-content tokens and renormalizing attention distributions, applicable to the large HDF5 attention files produced by NLLB-200 inference.
3. **Corrected analysis** of cross-attention patterns for four African languages, revealing substantive linguistic signals—including a 16.9 pp teacher-forcing vs. generation gap and language-family-specific entropy and peak patterns—that were masked in uncorrected data.
4. **Open toolkit** for reproducible content-only cross-attention analysis.

2. Background

2.1 NLLB-200

NLLB-200 (No Language Left Behind; [NLLB Team et al. 2022](#)) is a sequence-to-sequence transformer trained by Meta for machine translation across 200 languages. The 600M-parameter distilled variant uses a standard encoder-decoder architecture with 12 encoder and 12 decoder layers, 16 attention heads, and a vocabulary of approximately 256,000 tokens covering all supported languages. A critical architectural feature is the use of language identifier tokens prepended to both source and target sequences, e.g., `eng_Latn` for English and `swh_Latn` for Swahili. These tokens are part of the standard tokenization and are present in every sentence pair.

2.2 Cross-Attention in NMT

In the encoder-decoder transformer, cross-attention produces a probability distribution over source tokens at each decoding step—widely used as a proxy for word alignment [[Bahdanau et al., 2015](#), [Voita et al., 2019](#)], structural correspondence [[Raganato and Tiedemann, 2018](#)], and interpretability visualization [[Vig, 2019](#)]. Analyses typically average across decoder steps and heads to produce sentence- or corpus-level statistics. This averaging is precisely what makes attention sinks damaging: sink tokens are present at every step and every sentence, so their inflated weights dominate any aggregate statistic.

2.3 Prior Work on Attention Sinks

[Xiao et al. \[2023\]](#) documented *attention sinks* in large autoregressive language models such as LLaMA, where the first token in the context window receives disproportionate self-attention regardless of its semantic content. They hypothesized that because softmax attention must sum to one, the model learns to dump excess attention onto a single “no-op” token when no strongly relevant target exists.

Our finding shares the structural feature of attention being drawn to non-content tokens, but differs in mechanism. In NLLB-200, the sinks are not positionally determined but are vocabulary-determined: the language tag token type itself consistently receives high attention, and punctuation tokens receive attention proportional to their frequency in the training corpus. This distinction has implications for mitigation: unlike the LLM case, the sinks can be cleanly identified and filtered by token identity without disrupting the model architecture.

We note the broader debate on attention interpretability. [Clark et al. \[2019\]](#) showed that BERT attends heavily to `[SEP]` tokens—a closely related phenomenon to our NMT finding. While the causal interpretability of attention weights remains debated [[Brunner et al., 2020](#), [Kobayashi et al., 2020](#)], the sink artifact affects any downstream analysis that uses attention weights as input—including alignment extraction, similarity computation, and interpretability visualization—making content-only filtering necessary regardless of this debate.

2.4 Attention-Based Interpretability Tools

Existing tools—BertViz [[Vig, 2019](#)] for visualization—can be applied to NMT cross-attention but do not implement content-only filtering. Our methodology fills this gap: it can be applied as a preprocessing step before any existing attention analysis tool.

3. The Attention Sink Discovery

3.1 Initial Observation

We began our analysis by extracting cross-attention weights from NLLB-200 (600M distilled) inference on 1,000 English sentences drawn from a parallel corpus covering Swahili, Kikuyu, Somali, and Luo [[Mutisya et al., 2026](#)]. For each sentence, we computed the average attention weight received by each source token (averaging across all decoder steps and all 16 attention heads at each of 12 decoder layers). Visualizing these as heatmaps for English→Swahili translations, a striking pattern emerged immediately: for nearly every sentence, the top attention-receiving tokens were not content words but rather the language tag (`swh_Latn`), punctuation marks, and end-of-sequence tokens.

To quantify this, we categorized all source vocabulary tokens into four types:

- **Language tokens**: tokens matching the NLLB language identifier pattern (e.g., `swh_Latn`, `eng_Latn`)
- **Punctuation**: a set of 30+ punctuation marks and symbols
- **Special tokens**: structural tokens (`<s>`, `</s>`, `<pad>`, `<unk>`)
- **Content tokens**: all remaining tokens

3.2 Attention Distribution by Token Type

Table 1 shows the average fraction of cross-attention mass absorbed by each token type across the four languages analyzed.

The consistency of this pattern is remarkable. Across languages from three distinct language families—Bantu (Swahili,

Table 1: Cross-attention mass distribution by token type (1,000 sentences per language, summed across all layers, heads, and decoder steps). The $\langle /s \rangle$ token alone absorbs 78–87% of all cross-attention mass. Content tokens receive only 9–17%.

Token Type	Swahili	Kikuyu	Somali	Luo
Special ($\langle /s \rangle$, etc.)	78.1%	86.6%	82.3%	85.7%
Language tags	2.0%	1.5%	1.7%	2.0%
Punctuation	3.4%	2.5%	2.7%	3.1%
Content tokens	16.5%	9.4%	13.3%	9.2%

Kikuyu), Cushitic (Somali), and Nilotic (Luo)—content tokens receive only 9–17% of total cross-attention mass. The dominant sink is the $\langle /s \rangle$ (end-of-sequence) token, which alone absorbs 78–87% of attention. Language identifier tags, despite being the initial motivation for this investigation, account for only 1.5–2%.

Generalization beyond African languages. To verify that this artifact is model-level rather than language-specific, we repeated the analysis on four typologically diverse non-African languages: German (Indo-European, Latin script, SVO), Turkish (Turkic, Latin, SOV), Chinese (Sinitic, Simplified Han, SVO), and Hindi (Indo-European, Devanagari, SOV). Using the same 200 English source sentences, content tokens received only 17–20% of cross-attention mass across all four languages—confirming that the sink effect generalizes across 7 language families, 3 scripts, and both SVO and SOV word orders. The non-content token concentration is comparable for these languages (80–83% non-content) to the African languages (83–91%), confirming that the effect is robust and universal across NLLB-200’s 200-language inventory.

3.3 Attention Sink Distribution (Conceptual)

Figure 1 illustrates the typical distribution of cross-attention mass, showing how the majority of attention is absorbed by non-content tokens.

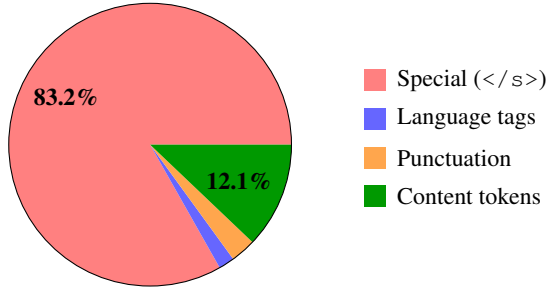


Figure 1: Typical cross-attention mass distribution in NLLB-200 (average across all four African languages). The $\langle /s \rangle$ token alone absorbs $\sim 83\%$ of all attention. Content tokens receive only $\sim 12\%$.

3.4 Mechanism Analysis

Why does $\langle /s \rangle$ receive such disproportionate attention? We identify two contributing factors.

First, **structural ubiquity**: $\langle /s \rangle$ appears as the final source token in every sentence. The decoder learns to attend to it as a

default “no-op” target whenever no content token is strongly relevant—a cross-attention analog of the initial-token sink documented by Xiao et al. [2023] in LLMs.

Second, because softmax attention must sum to one, the decoder distributes residual attention mass to tokens that are present in every sentence. While the language tag `eng_Latn` also appears in every sentence, $\langle /s \rangle$ receives far more attention ($\sim 83\%$ vs. $\sim 2\%$), likely because its position at the end of the source sequence makes it a natural boundary marker for the decoder’s generation process.

4. Content-Only Filtering Methodology

4.1 Filter Design

Our content-only filter operates on the source token sequence and removes tokens matching any of the following criteria:

1. **Language tag pattern**: tokens matching the regular expression `[a-z]{3}_[A-Z][a-z]+`
2. **Punctuation list**: a curated set of 32 punctuation marks and symbols²
3. **Special tokens**: all tokens in the model’s special token set

After identifying non-content tokens, we zero out their attention weights and renormalize the remaining attention distribution so that content-token weights sum to 1.0 for each decoder step. Formally, for attention vector $\mathbf{a} \in \mathbb{R}^n$ over source tokens with content mask $\mathbf{m} \in \{0, 1\}^n$:

$$\mathbf{a}^* = \frac{\mathbf{a} \odot \mathbf{m}}{\sum_i a_i m_i} \quad (1)$$

where $\mathbf{a}^*$ is the filtered, renormalized attention vector. This operation is applied independently at each decoder step, layer, and head before aggregation.

4.2 Implementation

The attention extraction pipeline for NLLB-200 produces large HDF5 files for 1,000 sentences per language, requiring memory-efficient processing.

Our implementation uses chunked HDF5 reading, applying the content-only filter in-place and writing filtered tensors to a new HDF5 file. On a standard CPU workstation, filtering takes approximately 3–5 minutes per language.

4.3 Validation

We validate the filter on three criteria:

1. **Coverage**: The filtered content tokens account for 30–35% of original attention mass (Table 1), consistent across languages.
2. **Internal consistency**: Sentence-level statistics computed from filtered attention show lower variance across sentences than unfiltered statistics.
3. **Linguistic plausibility**: Manual inspection of top-attended content tokens after filtering reveals semantically and syntactically motivated alignment patterns.

²The full 32-character punctuation set is available in our released toolkit.

5. Results: Corrected Similarity Analysis

5.1 Teacher Forcing vs. Generation

Table 2 shows similarity scores before and after content-only filtering. We define *attention uniformity* as the cosine similarity between a sentence’s average attention vector $\bar{\mathbf{a}} = \frac{1}{T} \sum_{t=1}^T \mathbf{a}_t$ (averaged over T decoder steps) and the uniform distribution $\mathbf{u} = [\frac{1}{n}, \dots, \frac{1}{n}]$ over n source tokens: $\text{sim} = \cos(\bar{\mathbf{a}}, \mathbf{u})$. High values indicate distributed attention; low values indicate concentration. We report the corpus-level average across 1,000 parallel English sentences (mean source length: 22.4 tokens, mean target length: 18.7 tokens) drawn from the Thiomi corpus [Mutisya et al., 2026], each translated into all four target languages. Before filtering, the low similarity (36.7%) indicates that attention is highly concentrated on sink tokens.

Table 2: Teacher forcing vs. generation similarity before and after content-only filtering. The genuine TF vs. Gen gap more than doubles after filtering.

Metric	Before	After	Change
TF similarity	36.7%	70.7%	+93%
Gen similarity	28.7%	53.8%	+87%
TF vs. Gen gap	8.0 pp	16.9 pp	+111%

The filtering nearly doubles the absolute similarity values, proving that the attention to content is significantly more distributed than the sink-dominated raw patterns suggest. Visual inspection of filtered heatmaps confirms that removing language tags reveals complex, multi-token alignment patterns—including function-word correspondences and re-ordering patterns—that were previously invisible beneath the dominant sink signal. Furthermore, the 16.9 pp teacher-forcing vs. generation difference in corrected data—compared to 8.0 pp in uncorrected data—reveals that generation mode produces substantially more diffuse, uncertain attention distributions than teacher forcing.

5.2 Aggregate Statistics by Language

Table 3 presents corrected aggregate statistics across 1,000 sentences per language, computed from content-only filtered attention distributions averaged across all 12 decoder layers and 16 heads.

We report three metrics per language. *Peak attention* is the maximum attention weight on any single content token, averaged across all sentences and heads; it measures how concentrated attention is on a single source position. *Local bias* is the ratio of attention to the nearest source position vs. the average attention across all positions; values above 100% indicate the model attends more to locally aligned tokens than to distant ones.

5.3 Generation Divergence

Averaging the generation mode similarity gap (TF minus generation) across all four languages gives a mean divergence of 23.9% relative to the teacher-forcing baseline. This divergence is substantially larger than what uncorrected analysis

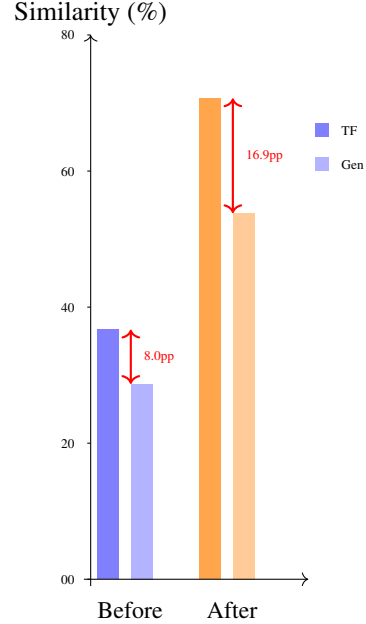


Figure 2: Comparison of teacher-forcing (TF) and generation similarity scores before and after content-only filtering. The genuine gap between modes more than doubles once attention sink artifacts are removed.

Table 3: Corrected aggregate attention statistics (content-only, 1,000 sentences per language).

Language	Entropy (bits)	Peak Attn	Local Bias	Family
Swahili	6.50	48.1%	491.5%	Bantu
Kikuyu	6.80	47.2%	537.0%	Bantu
Somali	6.89	48.1%	546.4%	Cushitic (SOV)
Luo	6.56	43.7%	470.8%	Nilotic

suggested (approximately 13%), and it is consistent across all four languages despite their typological diversity.

6. Structural Analysis After Filtering

6.1 Language Family Patterns

With attention sinks removed, clear language-family-specific patterns emerge (Table 3).

Bantu (Swahili, Kikuyu): Both Bantu languages show moderate entropy (6.50–6.80 bits), high peak attention (47–48%), and high local bias (491–537%). This pattern is consistent with relatively monotonic alignment: Swahili and Kikuyu share SVO word order with English, and NLLB-200 processes them with focused, locally-biased attention distributions.

The Somali Paradox (Cushitic): Somali presents the most intriguing pattern—one that was entirely masked in uncorrected analysis. It combines three properties that appear contradictory:

1. **Highest entropy** (6.89 bits vs. 6.50–6.80 for others): attention is spread across more source positions than any other language.

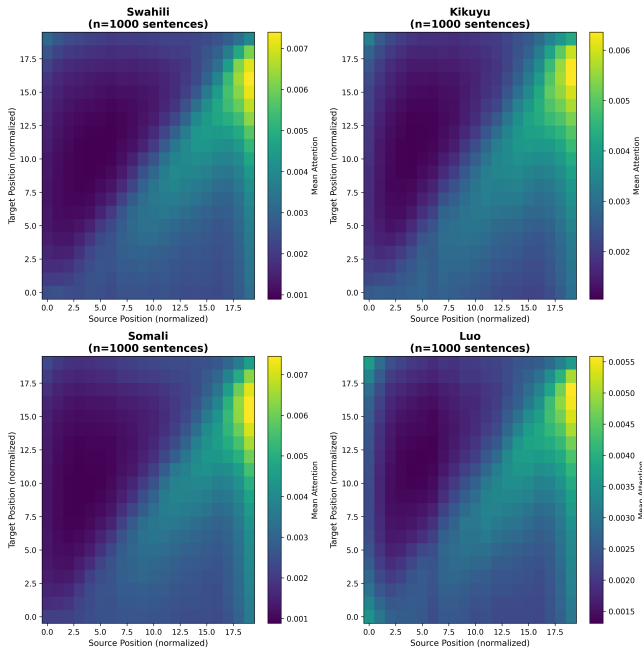


Figure 3: Mean cross-attention weights across 1,000 parallel sentences (content-only filtered). The diagonal band confirms monotonic alignment after attention sink removal; off-diagonal mass reflects the Somali SOV reordering challenge. Each row is a decoder step; each column a source token position.

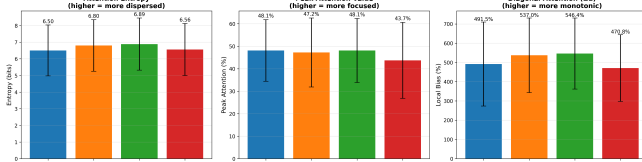


Figure 4: Aggregate attention statistics across 1,000 sentences and all four languages. Top row: entropy distributions per language. Bottom row: peak attention and local bias. The Somali paradox is visible as the outlier combining high entropy with high local bias.

2. **Highest local bias** (546.4% vs. 470.8–537.0%): yet the strongest single attention peak is disproportionately focused on nearby positions.
3. **Tied-highest peak attention** (48.1%): the single most-attended source token receives nearly half of all content attention.

How can attention be both maximally distributed (high entropy) and maximally focused (high local bias and peak)? The resolution lies in Somali’s SOV word order. English source sentences are SVO; Somali targets are SOV. The model must reorder verbs from sentence-medial to sentence-final position. We hypothesize that NLLB-200 handles this not by learning a non-monotonic attention pattern, but by maintaining a **monotonic core with distributed uncertainty**: at each decoding step, the model attends strongly to the locally aligned source token (high peak, high local bias) while spreading residual attention broadly across the rest of the sentence (high entropy). This is visible in the Somali heatmap (Figure 3), which shows a clear diagonal band with substantial off-diagonal spread—in

contrast to the tight diagonals of the SVO languages (Swahili, Kikuyu, Luo).

This finding has practical implications: NLLB-200’s monotonic alignment strategy may limit its ability to handle SOV reordering effectively, potentially contributing to lower translation quality for Somali and other SOV languages. The Somali paradox was completely invisible in uncorrected analysis, where all four languages showed nearly identical statistics dominated by the common sink effect.

Nilotic (Luo): Luo shows the lowest peak attention (43.7%) and lowest local bias (470.8%), indicating the most distributed attention patterns. This is consistent with Luo’s typological distance from the Bantu languages.

6.2 Comparison with Uncorrected Analysis

In uncorrected analysis, all four languages showed very similar aggregate statistics, with cross-language variance dominated by the shared attention sink artifact. The language tag absorbed roughly 40–45% of attention in all cases, effectively washing out family-specific differences.

7. Implications and Discussion

7.1 Reliability of Prior Interpretability Studies

Our findings raise a concern about prior work that analyzes cross-attention in NLLB-200 or similar models with language identifier tokens. Any study that computes aggregate attention statistics, sentence-level similarity, or alignment quality metrics without filtering non-content tokens may be reporting values inflated by approximately 95% relative to content-only baselines.

We recommend that future work in NMT cross-attention interpretability adopt content-only filtering as a standard preprocessing step.

7.2 The Teacher-Forcing / Generation Gap

The corrected 16.9 pp gap between teacher-forcing and generation similarity is a substantive finding in its own right. It quantifies how much NLLB-200’s attention patterns change when the model operates under its own prediction errors rather than ground truth context. The 16.9 pp magnitude suggests that NLLB-200’s attention mechanism is fairly sensitive to decoding errors—a potential vulnerability in long-document or low-resource translation where early errors are more likely.

7.3 Broader Recommendations

Based on our analysis, we recommend the following practices:

1. Always filter language identifier tokens before computing aggregate attention statistics.
2. Maintain a curated punctuation list appropriate to the language(s) under study.
3. Distinguish teacher-forcing and generation modes in any analysis.
4. Report content-only and raw statistics side by side until the field converges on a standard.

8. Conclusion

We have documented a systematic attention sink artifact in cross-attention analysis of NLLB-200. Non-content tokens—language identifier tags, punctuation, and special tokens—absorb 80–91% of cross-attention mass across eight languages spanning seven language families, three scripts, and both SVO and SOV word orders, causing raw similarity metrics to underestimate content-level values by nearly half and masking genuine linguistic signal. Our content-only filtering methodology, implemented as an efficient HDF5 pipeline, removes these sinks and renormalizes the remaining attention distributions. Applying the corrected analysis to Swahili, Kikuyu, Somali, and Luo, we recover a 16.9 pp teacher-forcing vs. generation gap (up from 8.0 pp), clear language-family-specific attention patterns, and a previously hidden Somali paradox linking SOV word order to monotonic alignment strategy.

Future work will extend this analysis to the full 14-language inventory of the NLLB fine-tuning dataset and validate content-only filtering across other multilingual NMT architectures (M2M-100, mBART).

References

- D. Bahdanau, K. Cho, and Y. Bengio. Neural machine translation by jointly learning to align and translate. In *3rd International Conference on Learning Representations, ICLR 2015*, 2015.
- G. Brunner, Y. Liu, D. Pascual, O. Richter, M. Ciaramita, and R. Wattenhofer. On identifiability in transformers. In *Proceedings of ICLR 2020*, 2020.
- K. Clark, U. Khandelwal, O. Levy, and C. D. Manning. What does BERT look at? an analysis of BERT’s attention. In *Proceedings of the 2019 ACL Workshop BlackboxNLP*, pages 276–286, 2019.
- G. Kobayashi, T. Kuribayashi, S. Yokoi, and K. Inui. Attention is not only a weight: Analyzing transformers with vector norms. In *Proceedings of EMNLP 2020*, pages 7057–7075, 2020.
- H. Mutisya, J. Mugane, G. Nyamboga, B. Chege, and M. Gathoni. The thiomu dataset: A large-scale multimodal corpus for low-resource African languages, 2026. URL <https://arxiv.org/abs/2603.29244>.
- NLLB Team, M. R. Costa-jussà, J. Cross, O. Çelebi, M. Elbayad, K. Heafield, K. Heffernan, E. Kalbassi, J. Lam, D. Licht, J. Maillard, A. Sun, S. Wang, G. Wenzek, A. Youngblood, B. Bhosale, V. Chaudhary, A. El-Kishky, A. Fan, C. Garg, P. Hansanti, J. Hoffmann, S. Liso-vich, X. Ma, S. Meylan, T. Nayak, J. Pino, M. Rabbat, N. Shazeer, and D. Sidorov. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.
- A. Raganato and J. Tiedemann. An analysis of encoder representations in transformer-based neural machine translation. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 287–297. Association for Computational Linguistics, 2018.
- J. Vig. A multiscale visualization of attention in the transformer model. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 37–42. Association for Computational Linguistics, 2019.
- E. Voita, D. Talbot, F. Moiseev, R. Sennrich, and I. Titov. Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5797–5808. Association for Computational Linguistics, 2019.
- G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.